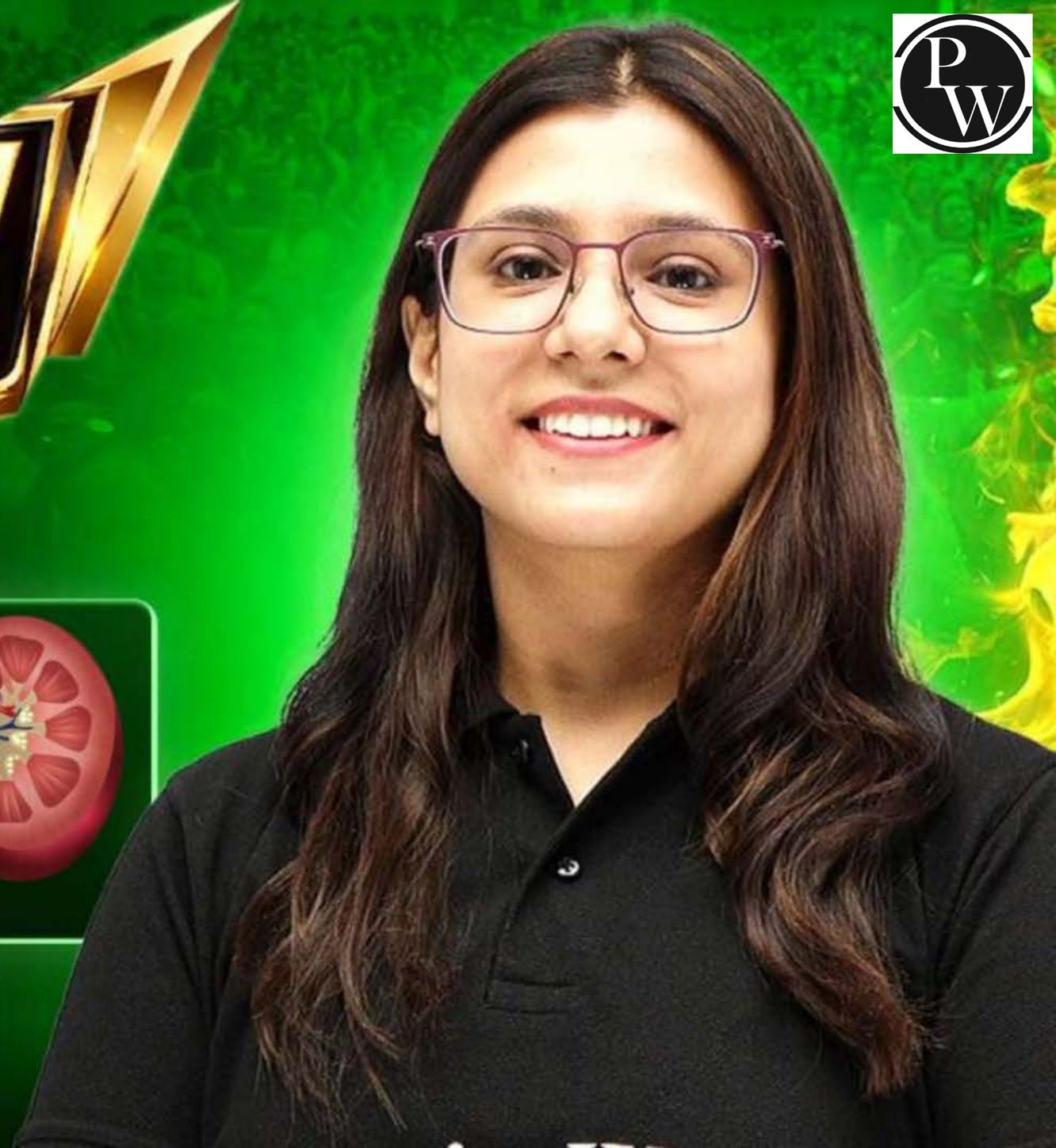
VICTORY 2027

The Excretory System



LECTURE 04 BIOLOGY

Upasana Ma’am



Topics To Be Covered

- Urine
- Abnormal Constituents in Urine
- Artificial Kidney
- Question


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

2027 Final

Nephron Diagram

↓

Labelling

Kidney Diagram



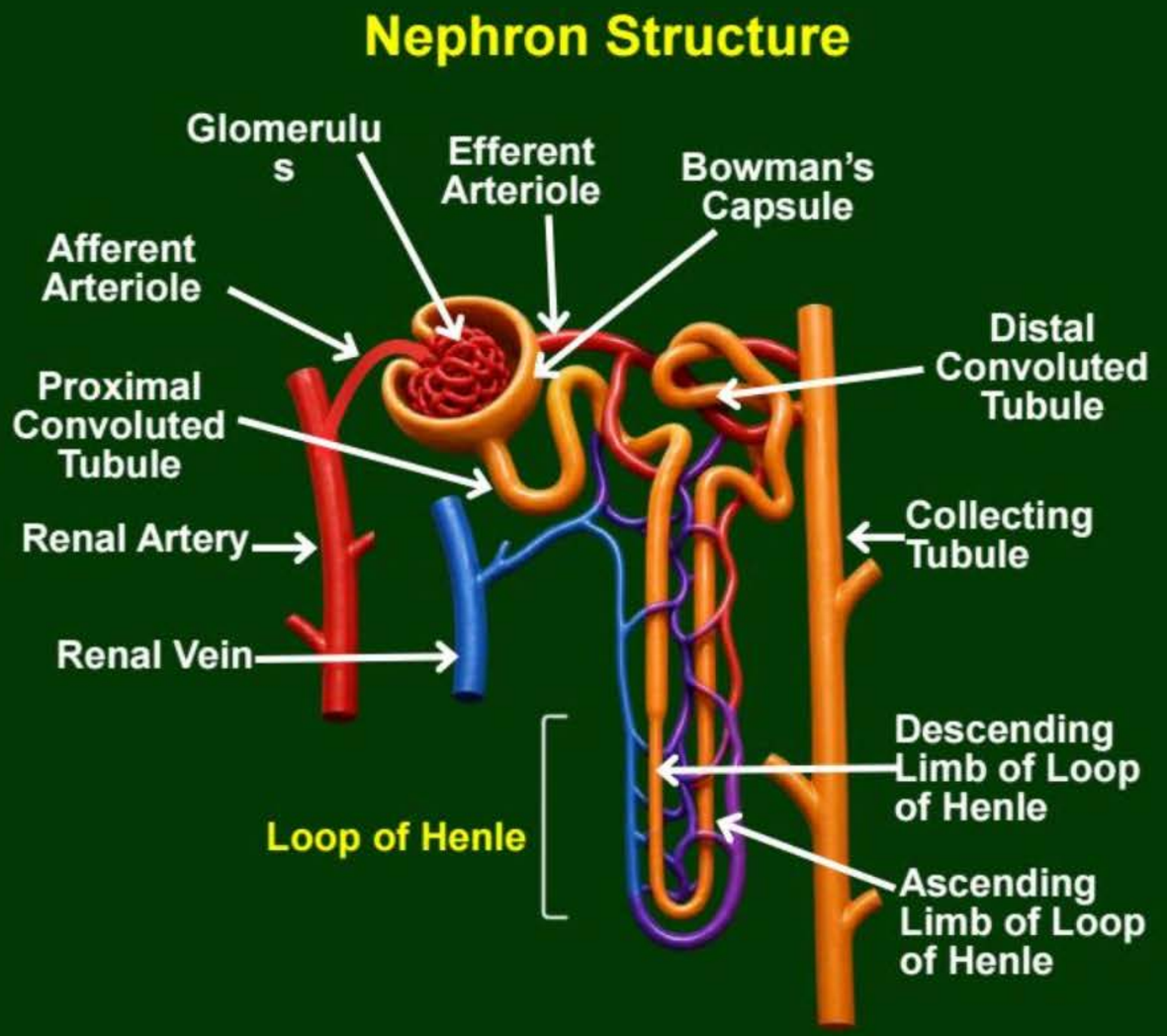
INTERNAL STRUCTURE OF THE KIDNEY

2 MAIN REGIONS

(ii) Inner lighter medulla

- Composed of finely striped substance arranged in several conical pyramids.
- Apex of each pyramid is called papilla which projects into the pelvis of kidney.
- Portions of conical tissue extend in between adjacent renal pyramids to form renal columns.
- Kidney is composed of an enormous number of minute tubules called uriniferous tubules or nephrons or renal tubules or just kidney tubules.

- Bowman's capsule is a thin-walled (single-cell thick epithelium) cup, something like a hollow ball pressed deep on one side. Its hollow internal space continues into the tubule. The outer concavity of the cup lodges a knot-like mass of blood capillaries, called glomerulus. The Bowman's capsule and the glomerulus together are called Malpighian capsule or renal capsule.



- Middle U-shaped part (Loop of Henle) is shaped like a hair-pin; it is not convoluted. It runs in medulla to turn back and to ?re-enter the cortex to continue into the next convoluted region of the tubule.

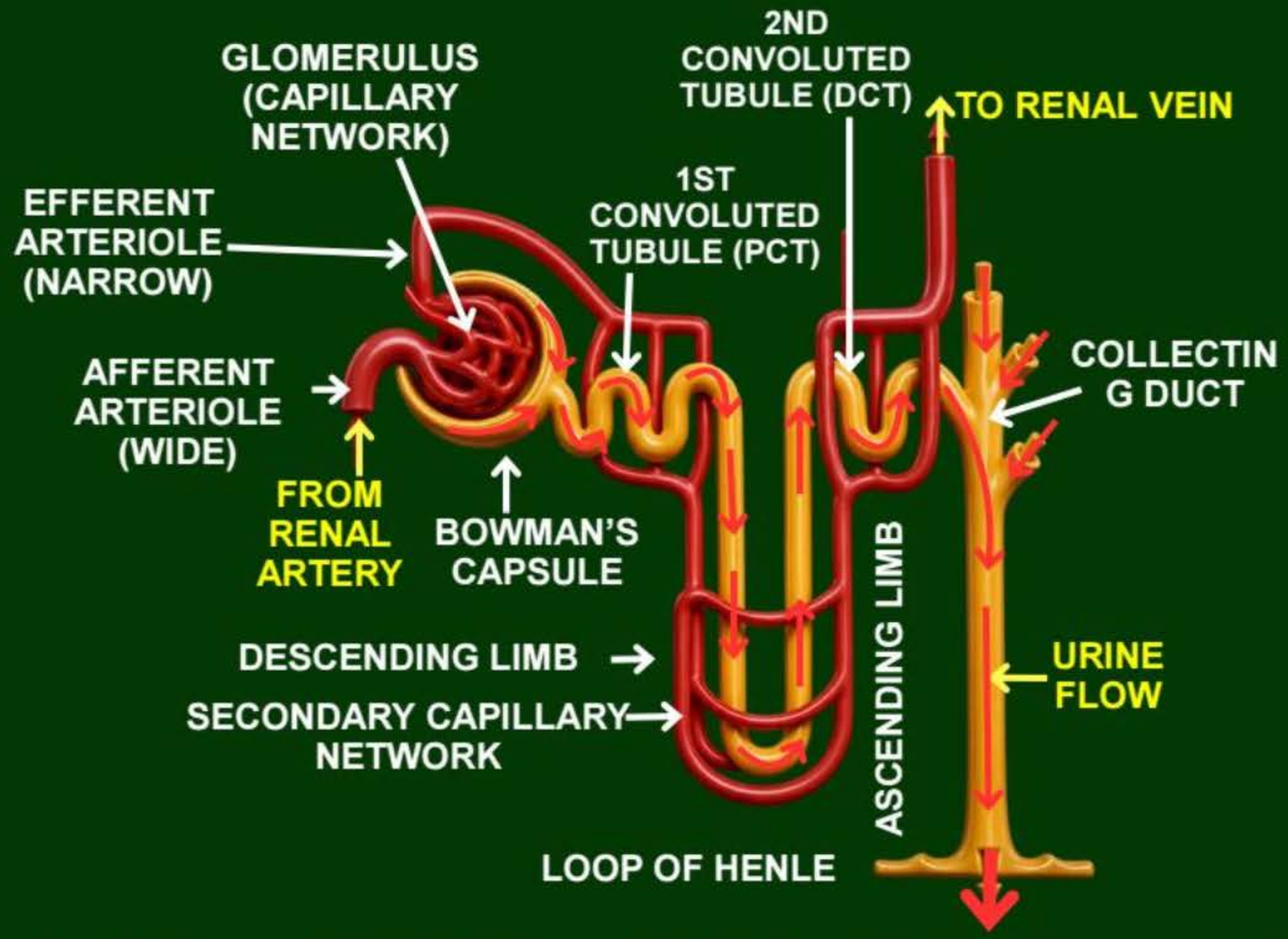


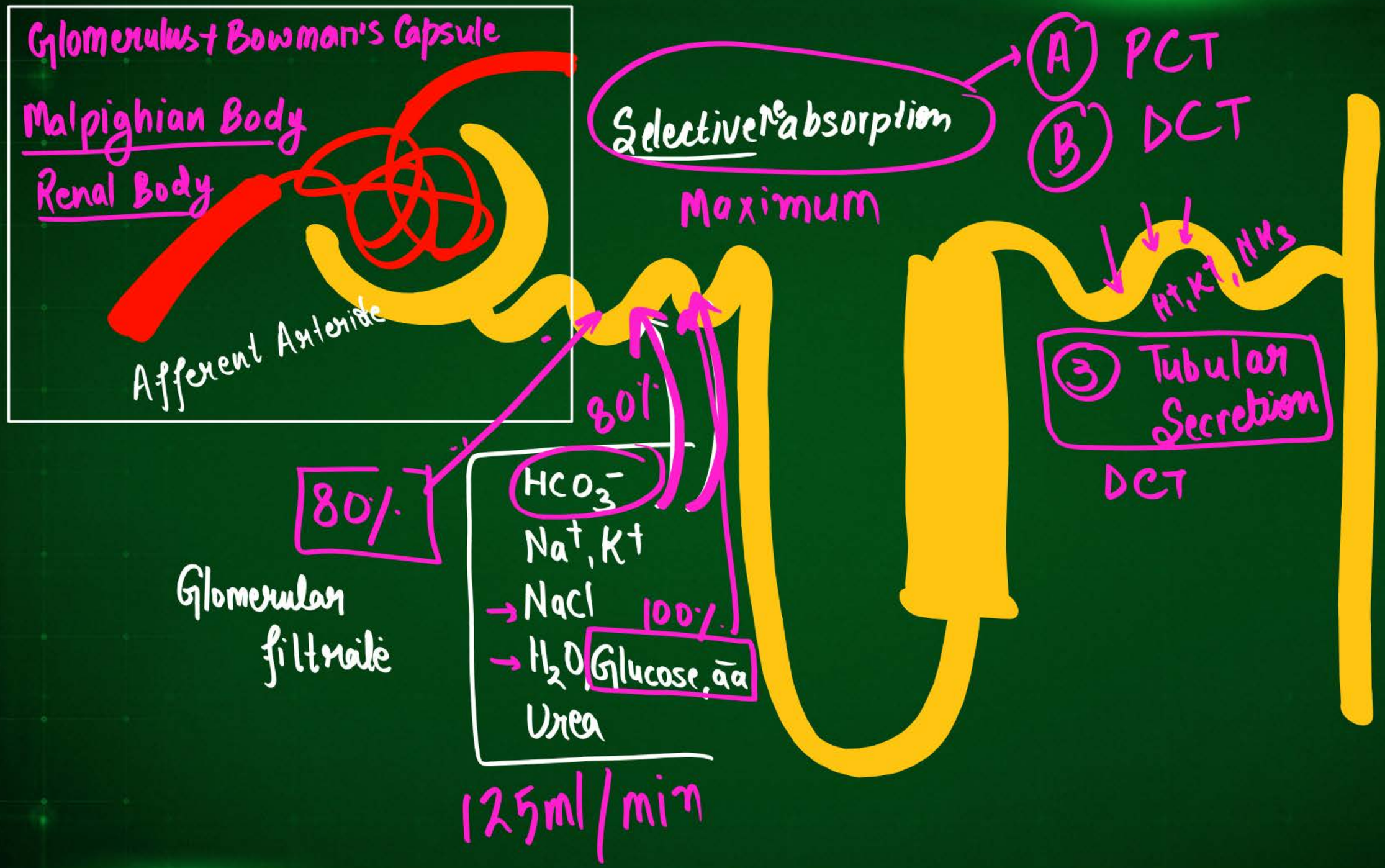
Fig. : A single renal (Uriniferous) tubule, highly diagrammatic



BLOOD SUPPLY TO THE KIDNEY TUBULES

1. Ultrafiltration :

- Part of blood plasma is filtered out under pressure through glomerulus into Bowman's capsule.
- The fluid entering the renal tubule is called glomerular filterate.
- Glomerular filtrate consists of water, urea, salts, glucose and other plasma solutes.
- Thus, the blood proceeding away from the glomerulus is relatively thick.


Glomerulus + Bowman's Capsule
Malpighian Body
Renal Body
Afferent Arteride
Selective reabsorption
A PCT
B DCT
Maximum
80%
80%
Glomerular filtrate
HCO3-
Na+, K+
→ NaCl
→ H2O
100%
Glucose, aa
Urea
125ml/min
H+, K+, NH3
3 Tubular Secretion
DCT

MARATHONS

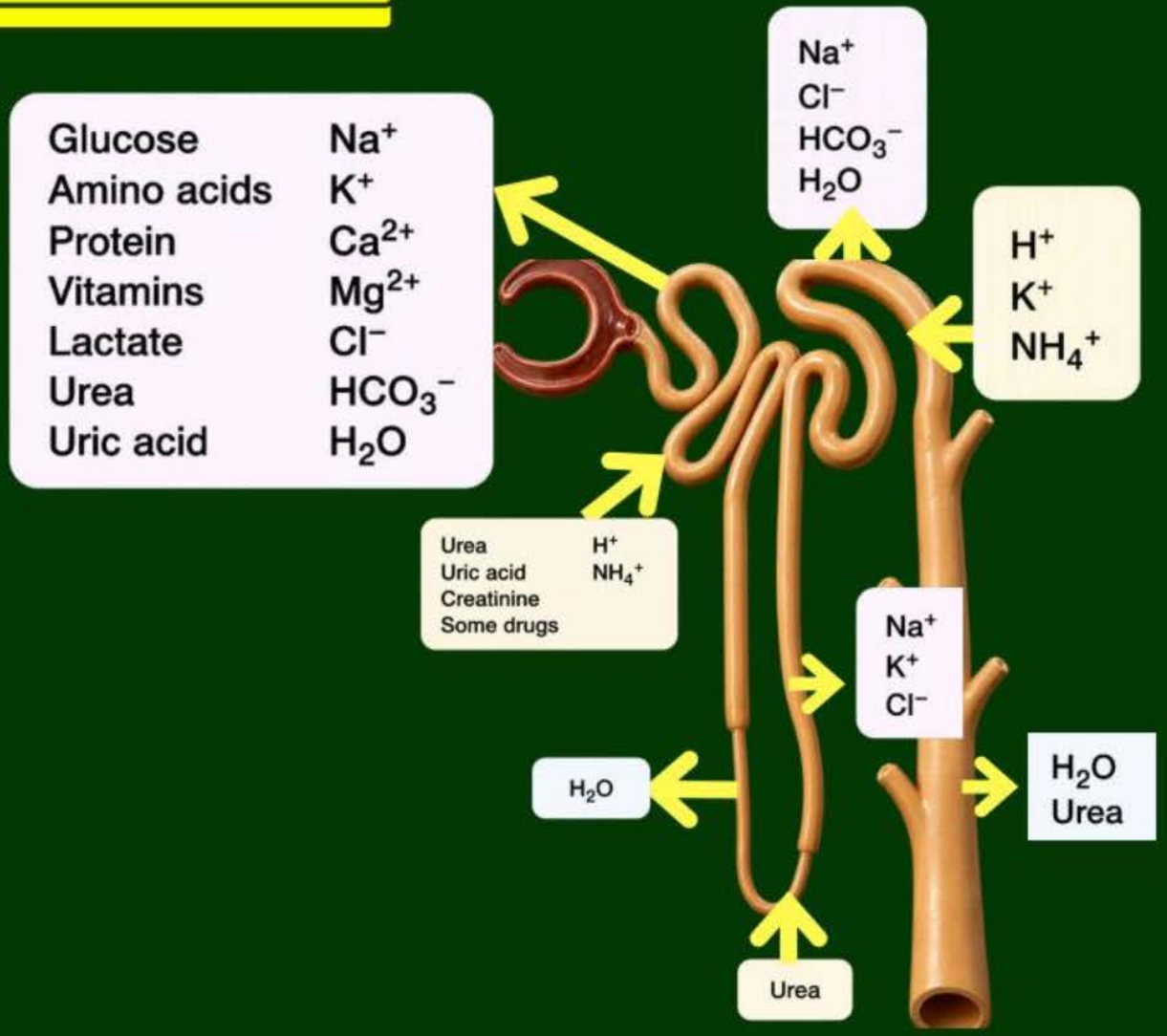
26th Aug. → 9th Sep

MID TERM BREAK

BLOOD SUPPLY TO THE KIDNEY TUBULES

2. Reabsorption :

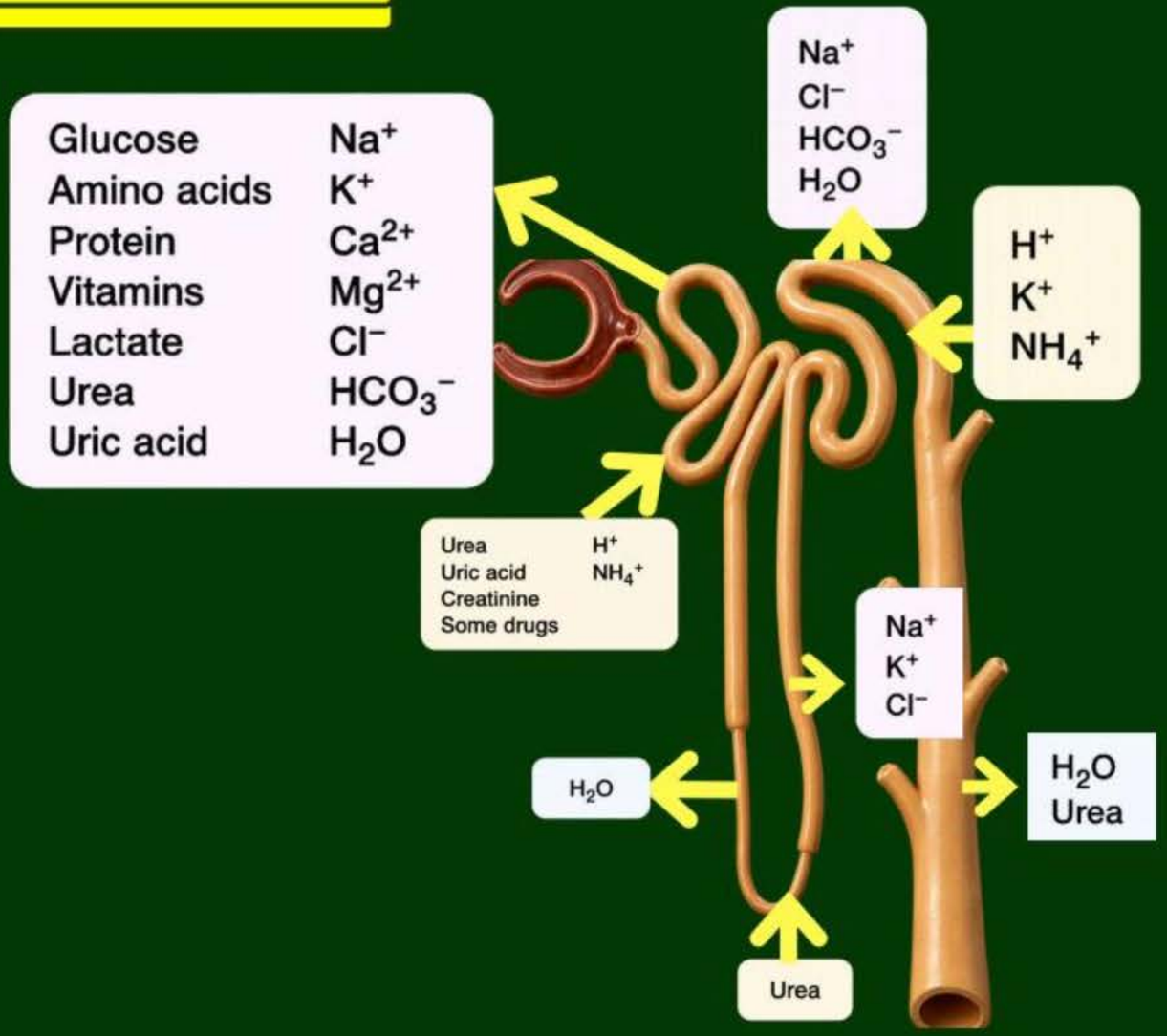
- Glomerular filterate entering the renal tubule is not urine.
- Selective absorption – reabsorption is only to the extent that the normal concentration of blood is not distributed.



BLOOD SUPPLY TO THE KIDNEY TUBULES

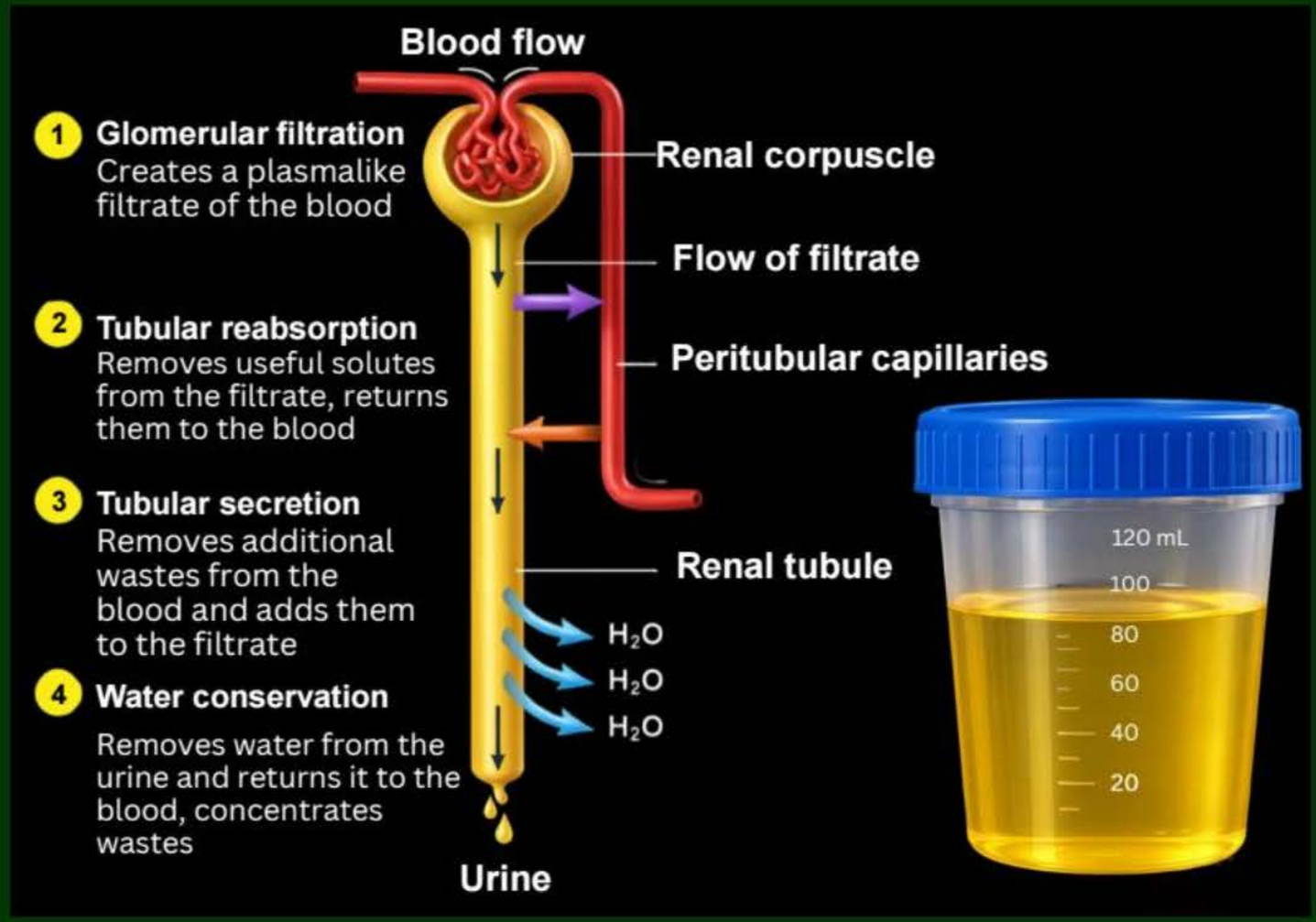
3.Tubular secretion :

- K^+ and chemicals like penicillin are passed into urine in DCT.
- This passage involves the activity of cells of tubular wall, hence called tubular secretion.
- The filterate left after reabsorption and tubular secretion is called urine.
- Oxygen demand of kidneys is 6 to 7 times higher than what is required by muscles.



URINE EXCRETION

- Urine is expelled from urinary bladder through urethra (penis male, directly in female) by relaxation of sphincter muscles located at opening of urethra under impulse from nervous system
- This process is called micturition.



MICTURITION



Motor Neuron

Signal from BRAIN

↓

Urinary Bladder

Sensory Neuron ✓

[signal from Urinary Bladder - BRAIN]

Sphincter Muscle.

Relax

Urine.

PHYSICAL PROPERTIES OF URINE

1. Colour – clear yellow , colour varies with diet

UROCHROME pigment [Urobilin]

2. Volume – 1 to 1.5 litres per day but varies

3. pH – 5 to 8, usually slightly acidic (pH = 6), protein diet makes it more acidic while vegetable diet makes it alkaline. (basic)

4. Odour – smell of urine becomes strong, ammonia-like due to bacterial activity, otherwise faint smell

5. Specific gravity – 1.003 to 1.035 ★

CONSTITUENTS OF URINE

- 95% of water
- 5% solid wastes

Organic in (g/L)

- Urea – 2.3 g/L
- Creatinine – 1.5 g/L
- Uric acid – 0.7 g/L
- Others – 2.6 g/L

CH–

($NH_2-CO-NH_2$)

waste

0.7 1.4 2.1



1 lit

CONSTITUENTS OF URINE

Inorganic in (g/L)

- Sodium chloride – 9 g/L NaCl
- Potassium chloride – 2.5 g/L KCl
- Ammonia – 0.6 g/L NH_3

ABNORMAL CONSTITUENTS IN URINE

Kidney stones → Renal Calculi

1. **Blood cells** – Known as haematuria

RBC → Hb

- Blood passes with urine due to some infection in urinary tract, kidney stones or tumour.



BROWN RED

ABNORMAL CONSTITUENTS IN URINE

2. **Glucose** – known as glycosuria

- Excess glucose passes with urine due to diabetes mellitus (sugar diabetes)

ABNORMAL CONSTITUENTS IN URINE

3. **Albumin** – Condition termed as albuminuria.

- Due to high blood pressure or increased permeability of Bowman's capsule membrane on account of bacterial infection.

- Albumin protein maybe passed via urine.

Albumin → a large sized protein, which normally does not come in Glomerular filtrate



reability of Bowman's Capsule.

ABNORMAL CONSTITUENTS IN URINE

4. **Bile pigments** – Due to anaemia, hepatitis (jaundice) or liver cirrhosis, urine may contain bile pigments.



low blood, low RBCs

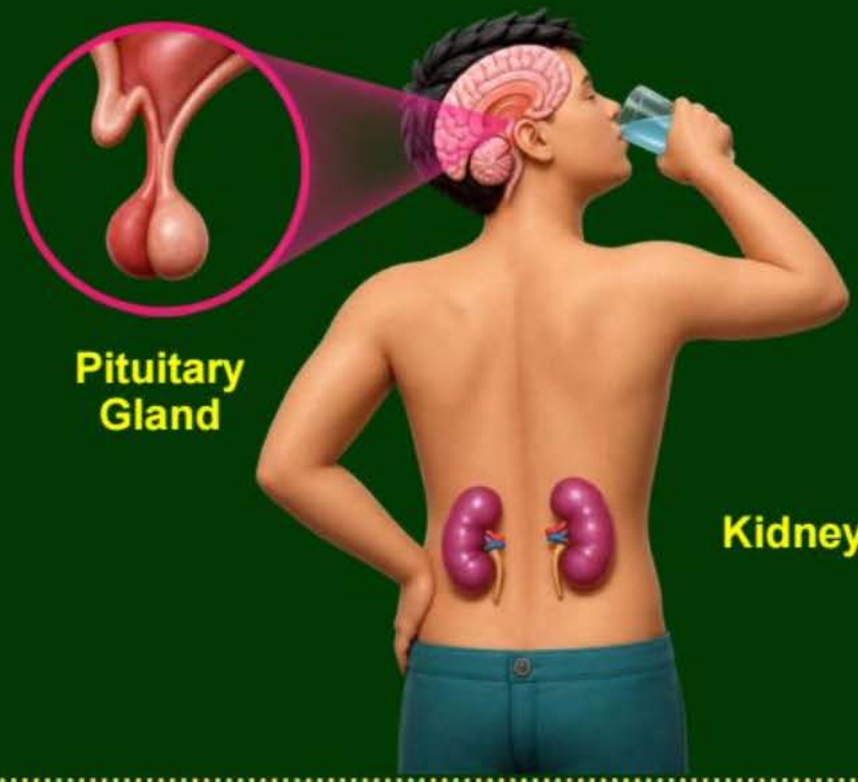
Adipose tissue (fat storing tissue)

FATTY LIVER → LIVER CIRRHOSIS

More Bile Pigments

DIABETES INSIPIDUS

- The condition of accumulation of urea in the blood termed as uremia
- Urine becomes tasteless when there is much water in urine
- Due to insufficient ADH hormone

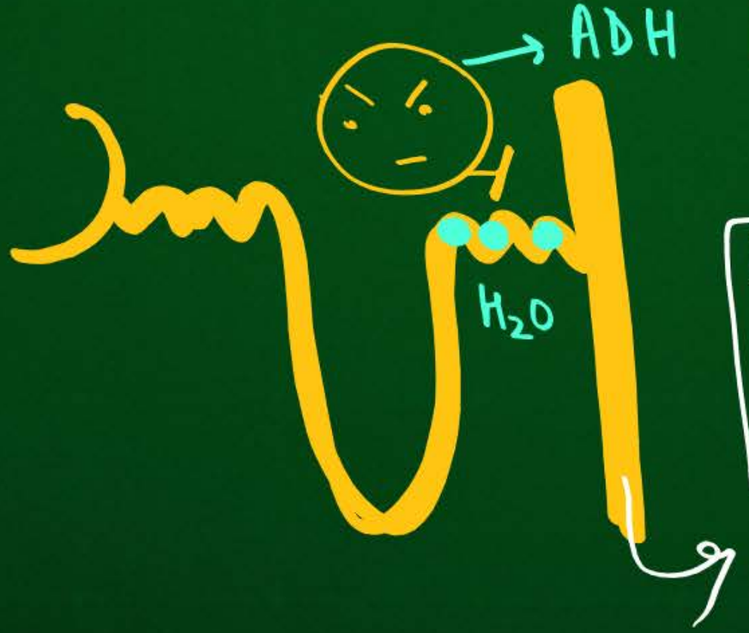


Pituitary gland produces insufficient ADH, hence the kidneys make a lot of urine.



coffee
Chai
Alcohol

Anti Diuretic Hormone (ADH) full form
OR
Vasopressin

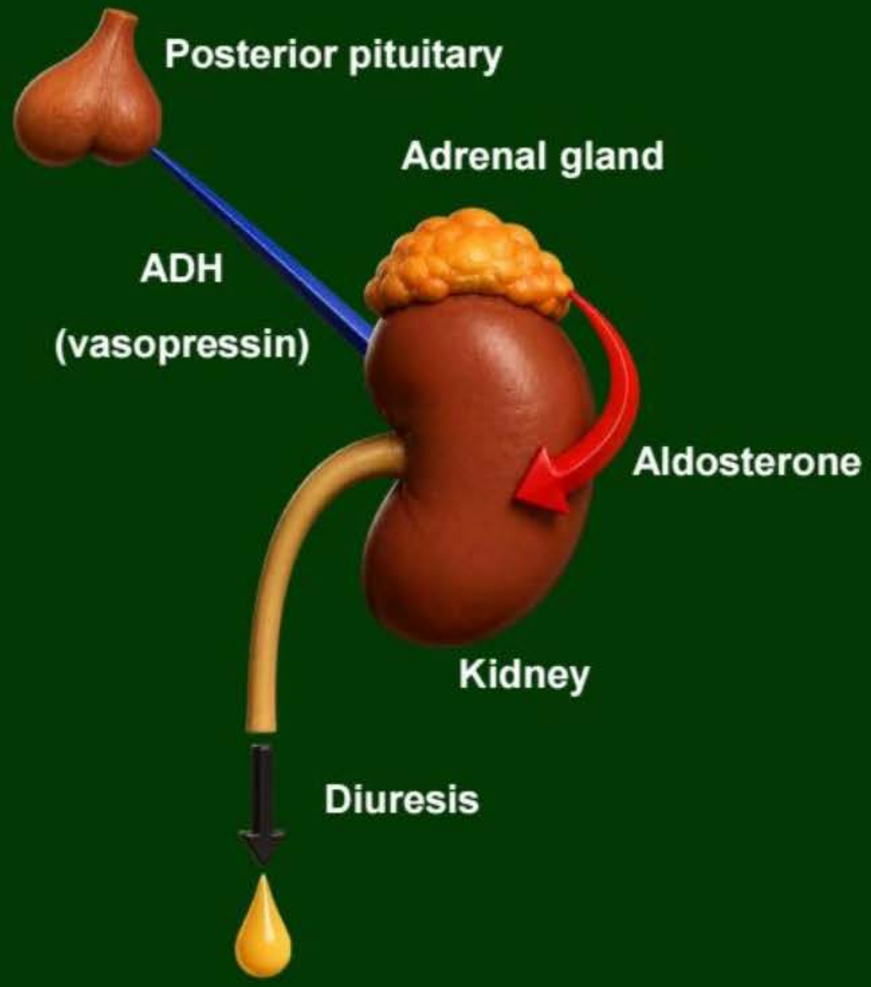


[To control the concentration of Water in urine]

After Diuretics
A) More H_2O in Urine
B) less " " "

REGULATION OF URINE OUTPUT

- Pituitary gland and diuresis (increased production of urine).
- Concentration of urine by water reabsorption is controlled by antidiuretic hormone (ADH) secreted by the posterior lobe of pituitary gland.
- If ADH secretion is reduced there is an increased production of urine known as “diuresis”.
- Liquid diets, tea, coffee, alcohol, etc causes increase in formation of urine.

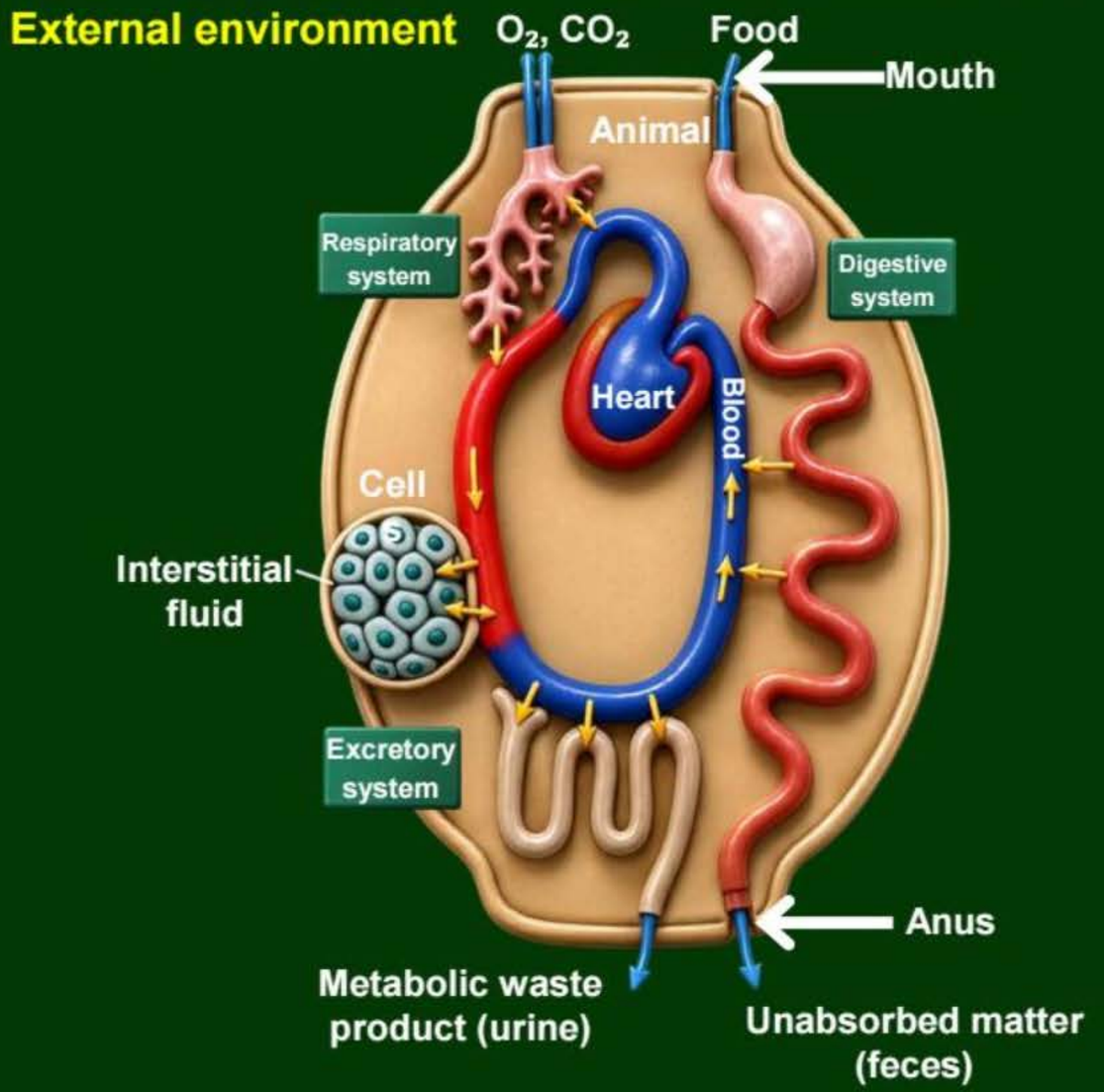


OSMOREGULATION

OSMOREGULATION

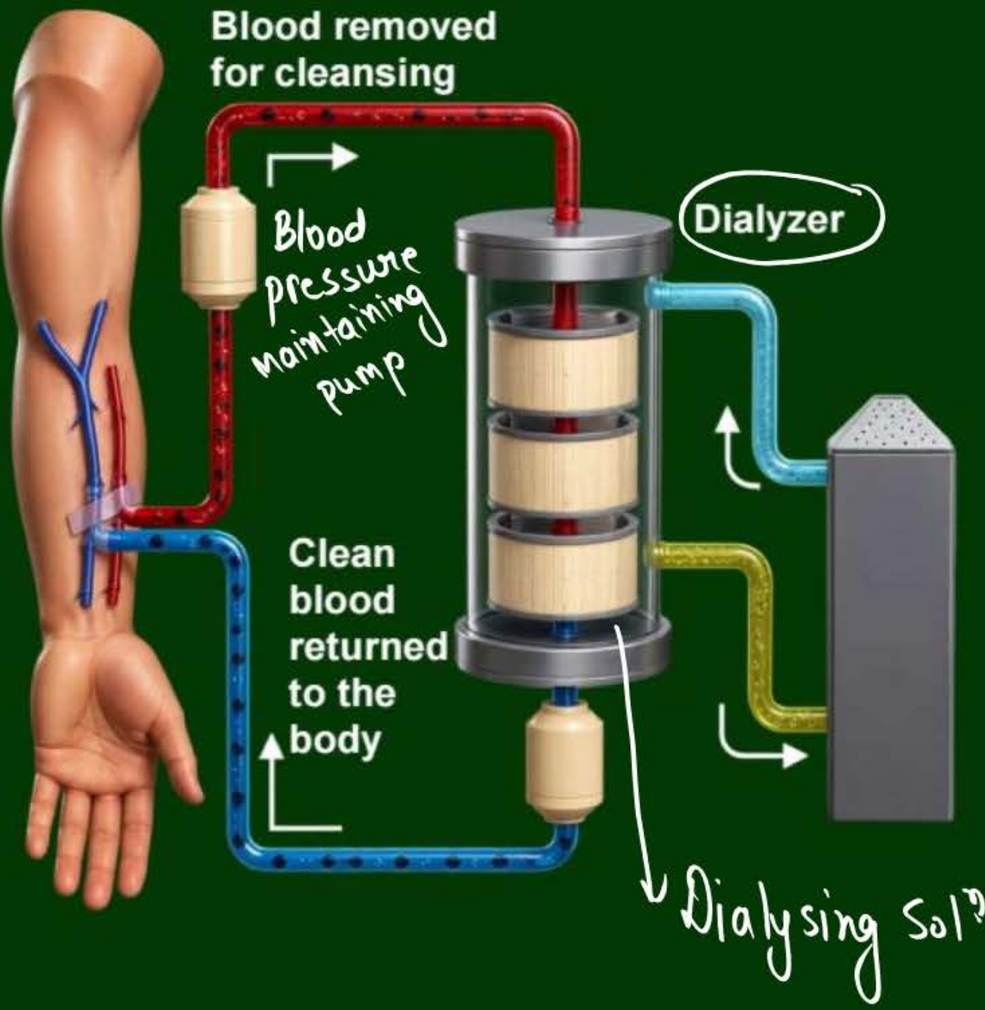
- Kidney while removing wastes like urea from the blood also regulates its composition, i.e., the percentage of water and salts.

- It implies the regulation of osmotic pressure of blood.



ARTIFICIAL KIDNEY

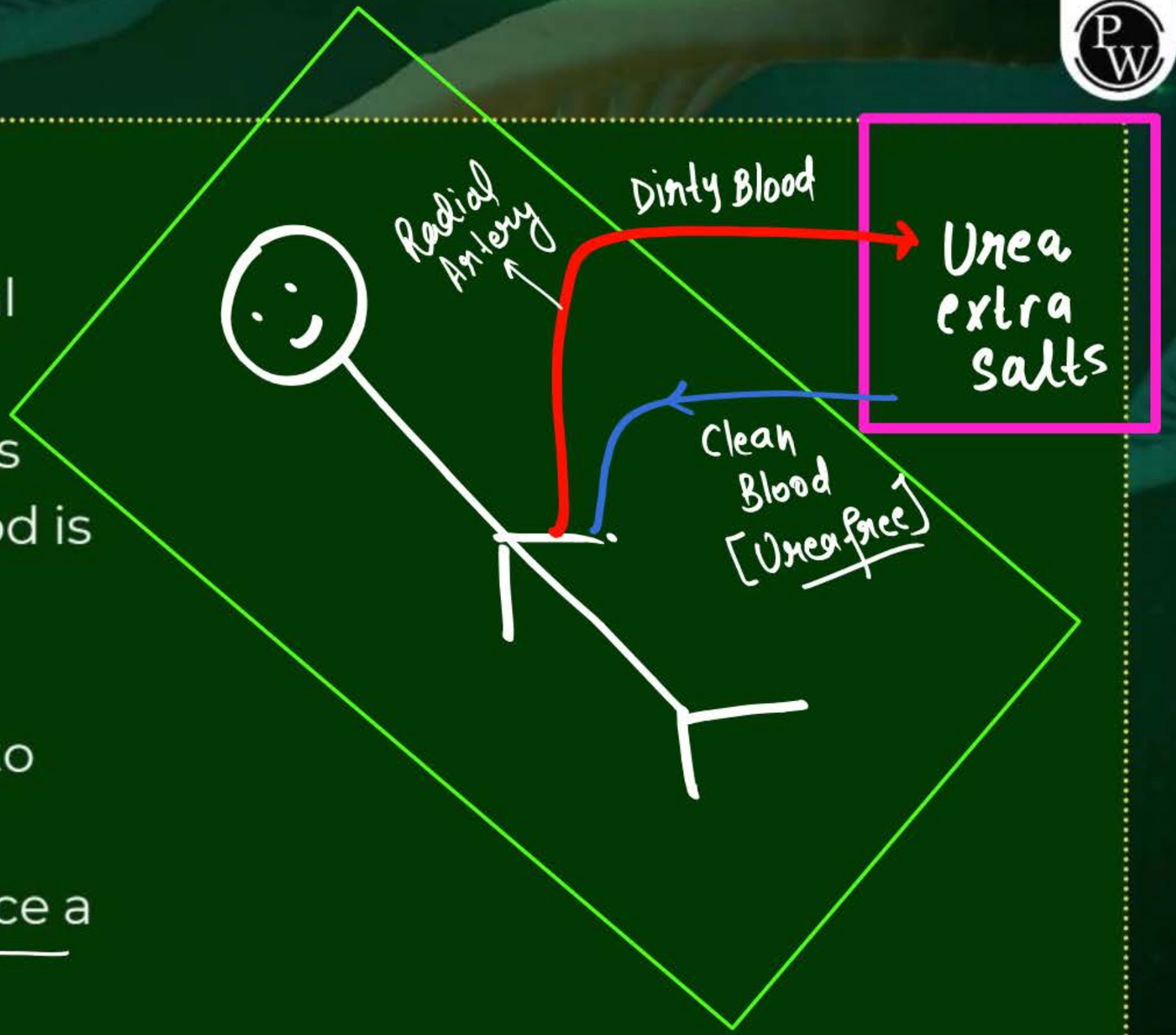
- If one kidney is damaged or removed, the other kidney is sufficient for excretory needs and person can lead to a normal life.
- But failure of both kidneys would lead to death.
- Hence an artificial kidney is a dialysis machine.



ARTIFICIAL KIDNEY

- Patient's blood is led from radial artery in his arm through the machine where urea and excess salts removed and purified blood is returned of vein in same arm.

- In case of permanent damage to the kidneys, dialysis is to be repeated for about 12 hours twice a week.



QUESTIONS

What is the condition of accumulation of urea in the blood termed as?

A renal calculi

B glomerulonephritis

C uremia

D ketonuria

QUESTIONS

Which one of the following is also known as an antidiuretic hormone?

A. Oxytocin

B. Vasopressin ✓

C. Adrenaline

D. Calcitonin

Duggu has **Cholera**

Is! excessive loss of H_2O from Body.

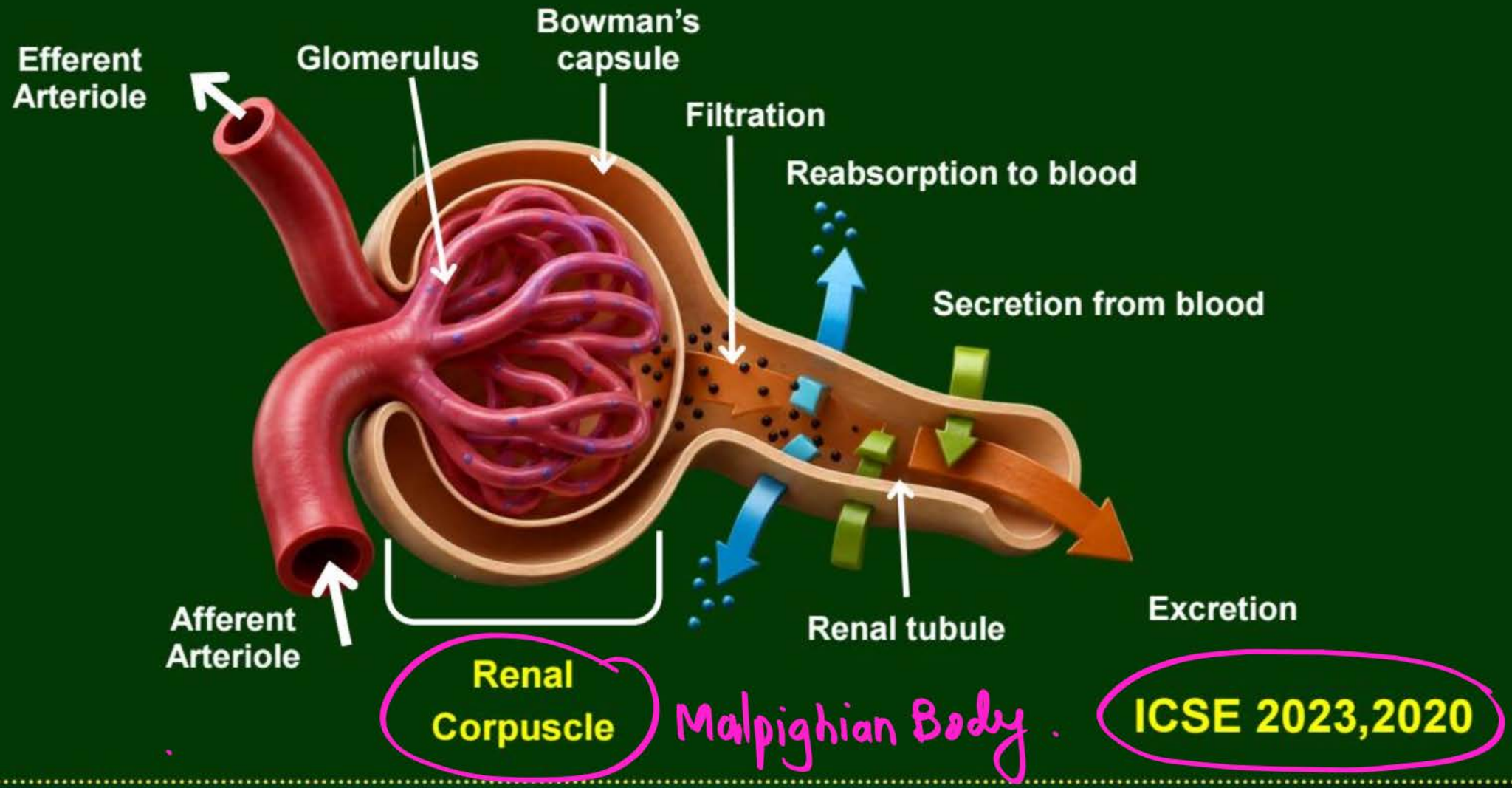
Now Kidney will not pass extra H_2O in urine.

↓

NO URINATION

It causes UREA accumulation in Body

[UREMIA] / [URAEMIA]


Efferent Arteriole
Glomerulus
Bowman's capsule
Filtration
Reabsorption to blood
Secretion from blood
Afferent Arteriole
Renal tubule
Excretion
Renal Corpuscle
Malpighian Body
ICSE 2023,2020

Haem → Bile Pigments

Biliverdin (green)

↓

Bilirubin (yellow) Waste.

Thank You